

	ADNOC GAS DBOOM WORKS FOR RENTAL COMPRESSION FACILITIES FOR BUHASA PLANT (PROJECT 5.0)				
	ADNOC DOCUMENT NO: DOD-30201-50150-BH-000-16-00-004 CONTRACTOR DOCUMENT NO: ADN-AEC-EL-DAT-009	Rev.: A			
DATASHEET - GAS POWER GENERATOR					
Sr. No.	DESCRIPTION	Project Requirements	Data Filled By Vendor	ASTRA Response 20241107	MOM 20241107
1.0	GENERAL			Vendor shall provided response in this sheet alongwith all the files shared with this email. Folder name "Pending Compliance" is not yet filled or responded by Vendor. Vendor has not submitted any catalog or the certificate or compliance. Emission Data is not yet responded. Synchronization scheme is not yet responded.	MOM as recorded in this file 20241107. Vendor Catalogs shall be submitted by 20241108. Emission data compliance confirmed with "Additional Catalytic convertor". Synchronization scheme compliance is confirmed.
1.1	Gas Generator Quantity	02	02	-	Total Gas Engine Generator required 02 number. Associated items shall be as per the standard supply for each item enslited in this Datasheet. E.g. Control Panels etc shall be dedicated for each generator. There is no common system/equipment between two generators.
1.2	Gas Generator Tags	G-4101-01A/B	KGW550 GF-T A/B	550kW Model noted. Project tags shall be used	-
1.3	Make & Model Number of Alternator	VTA	LEROY SOMER&LSA 50.2 V10/6P	-	
1.4	Country of Origin of Alternator	VTA	CHINA	-	
1.5	Make & Model Number of Engine	VTA	Waukesha&F3524GSI S5	-	
1.6	Country of Origin of Engine	VTA	America	-	Canada
1.7	Acoustic & Weatherproof enclosure	Required	Confirm	-	
1.8	CO2 Fire Suppressing System as per NFPA	Required	Confirm	-	
1.9	Design Life	06 years	Confirm	-	
1.10	International Codes and Standards [Refer: Attachment-1]	Compliance Required As Applicable	Confirm	-	
1.11	Vendor Deviation Form [Refer Attachment-2]	Vendor shall fill in the attached form in case of any deviation	Confirm	-	
1.12	Site Conditions [Refere Attachment-3]	Compliance Required As Applicable	Confirm	-	
1.13	Vendor Documents List [Refer Attachment-4]	Compliance Required	Confirm	-	

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1.14	Gas Gensets Operation: Parallel Generator Control Relay shall have provision of manual and automatic switching from one unit to the other after defined time	Compliance Required	Confirm	-	
2.0	SITE & INSTALLATION CONDITION				
2.1	Area of Installation (Indoor/Outdoor)	Outdoor	Outdoor	-	
2.2	Area of Classification (Hazardous / SafeArea)	Safe Area	Safe Area	-	
2.3	Design Ambient Temperature - Maximum	58 Deg C	55Deg C	Compliance to 58 is mandatory Process: Please state the reason specifically why 58C cannot be considered?	For Generator inside container temperature upto 58 Deg C still confirm the full rated output under the ambient outside temperature at 55 Deg C.
2.4	Design Ambient Temperature - Minimum	05 Deg C	05 Deg C	-	
2.5	Altitude Above Mean Sea Level	Less than 1000 meters	Less than 1000 meters	-	
2.6	Ambient Conditions	Dusty & humid	Dusty & humid	-	
3.0	DESIGN PARAMETERS FOR ENGINE				
3.1	Type of Design	VTA	4-cycle and turbocharging	-	
3.2	Applicable Standard	ISO	ISO	-	
3.3	Rated Continuous Power at Site Ambient Temperature (kW)	525 kW @ 0.8 pf	525 kW @ 0.8 pf	-	
3.4	Rated Speed (RPM)	VTA	1000	-	
3.5	Number of Strokes	VTA	4	-	
3.6	Direction of Rotation from Drive End	CCW	CCW	-	
3.7	Overload Capability	110% of Rated Power	110% of Rated Power	-	
3.8	Exhaust Gas temperature:				
3.9	a) At outlet of cylinder	VTA	622Deg C	-	
3.10	b) Upstream turbo charger	VTA	587Deg C	-	
3.11	Cooling System	Air Cooled Horizontal Radiator or alternate as per Vendor Design	Air Cooled Horizontal Radiator	-	
3.12	Type of Starting Mechanism	Electric Start with Battery Backup	Electric Start with Battery Backup	-	
3.13	Maximum Noise Level measured at 1 Meter distance from the enclosure	85 dB (A)	85 dB (A)	-	
3.14	Time Required for Start from Cold Conditions	VTA	10S	-	
3.15	Type of Fuel	Fuel Gas H2S Content: 50 ppm	Confirm Attachment-1	Vendor Response awaited	Emission data compliance confirmed with "Additional Catalytic convertor" and input gas 50ppm H2S
3.16	Fuel Consumption at Rated Voltage	VTA	141Nm3/h@525 kW	-	

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3.17	Cylinder Arrangement	VTA	6/L	-	
3.18	Bore, mm	VTA	238	-	
3.19	Compression Ratio	VTA	9.7:1	-	
3.20	Instruments: Orifice Type Fuel Gas Metering Instrument [Separate for each unit] Lube Oil Hight Temperature Over Speed Vibration Lube Oil Temperature Others As per Vendor Standard Design & Monitoring and Protection	Required	Instruments: Orifice Type Fuel Gas Metering Instrument [Separate for each unit] Lube Oil High Temperature Over Speed Vibration Lube Oil Temperature Others As per Vendor Standard Design & Monitoring and Protection	-	
3.21	Control: Mechanical Tachometer Lube Oil Pressure Gauge Water Temperature Gauge Running Manifold Vaccum Gauge Running Time Meter Start/Stop Control Exhaust Gas Temperature Cooling Water Pressure Others As per Vendor Standard Design	Required	Speed, Lube oil pressure, Water temperature, and Running Manifold Vaccum parameters are displayed in the generator set control panel,so there is no need to add additional gauges.	-	
3.22	All measurements and control shall be provided to the DCS through redundant RS485 communication link	Required	Confirm	-	
3.23	Lube Oil System	Required	Confirm	-	
3.24	Engine Exhaust System with Silencer	Required	Confirm	-	
3.25	Shaft Couplings	Required	Confirm	-	
3.26	Torsional Analysis of Engine	Required	Confirm	-	
4.0	FUEL GAS				
4.1	Fuel Gas Pressure	7 barg	Confirm	-	
4.2	Min / Max Pressure	VTA	2.76-4.14barg	-	
4.3	Size of fuel line	VTA	2"	-	
4.4	Emergency push button	VTA	Confirm	-	
4.5	Lubrication consumption	VTA	0.15g/kwh	-	

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4.6	Maximum Number of Continuous Operating Hours without maintenance interval (Hours)	VTA	4000h	-	
4.7	Duration of Shutdown for Routine maintenance after above Interval	VTA	16h	-	
4.8	Starting type	VTA	Electric starting	-	
4.9	NOx after SCR (selective catalytic reduction treatment	As per Scope of Work	72mg/Nm3@3% O2	Confirm inclusion of the SCR system and monitoring system. Please confirm that 72 mg/Nm3 is on dry basis.	Confirmed
4.10	Exhaust Gas Treatment System	Incompliance to Sox Nox Project Specification, if required	Compliance to Sox Nox Project Specification.	Confirm that the system is included in the offer. Please confirm SO2<35mg/Nm3.	SO2<35mg/Nm3 is confirmed as Engine outlet. Further additional "Exhaust Gas Treatment System" is not required.
5.0	DESIGN PARAMETERS - ALTERNATORS				
5.1	Rating capacity	525 kW@ 58 Deg C	550 kW@ 55 Deg C	Please state the reason specifically why 58C cannot be considered?	For Generator inside container temperature upto 58 Deg C still confirm the full rated output under the ambient outside temperature at 55 Deg C.
5.2	Rated Voltage (V)	415V	415	-	
5.3	Rated current	VTA	957	-	
5.4	Number of Phases	3P+4W	3P+4W	-	
5.5	Rated Power Factor	0.8	0.8	-	
5.6	Rated Frequency (Hz)	50	50	-	
5.7	Rated Speed	VTA	1000	-	
5.8	Stator Connection	Star Connection	Star Connection	-	
5.9	Generator Duty	S1	S1	-	
5.10	Rotation facing Drive End	VTA	Clockwise	-	
5.11	Winding Insulation- Generator & Exciter	Class H	Class H	-	
5.12	Protection method	IP 23	IP 23	-	
5.13	Temperature rise	Class F	Class F	-	
5.14	Overload Capability	VTA	10%	-	
5.15	Generator Efficiency (%) - at unity powerfactor:	VTA	95%	-	
5.16	3 Phase short circuit withstand time	3s	3s	-	
5.17	Recovery time to rated voltage	VTA	5s	-	
5.18	Speed control	Electronic Governor	Electronic Governor		
5.19	Type of Excitation Self excitation, Brushless type, consisting of a main AC exciter and a permanent magnet generator direct coupled and air cooled	Required	Confirm	-	

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5.20	Excitation - Rated Current At least 110% of the excitation current at the rated output of the generator under the most stringent operating conditions	Required	Confirm	-	
5.21	Excitation - Rated Voltage At least 110% of the rotor voltage at the rated output of the generator under the most stringent conditions	Required	Confirm	-	
5.22	Winding temperature detectors (No & type)	2 RTD per Phase, PT-100 platinum resistance elements as per IEC	Confirm	-	
5.23	Bearing temperature detectors (No & type)	2 RTD per Phase, PT-100 platinum resistance elements as per IEC	Confirm	-	
5.24	Common Engine / Generator control panel	Required	Confirm	-	
5.25	AMF [Auto Main Failure] Panel	Integrated with Control Panel or Separate as per Vendor standard design	Confirm	-	
5.26	Design Temperature (°C)	58 Deg C	55 Deg C	Please state the reason specifically why 58C cannot be considered?	For Generator inside container temperature upto 58 Deg C still confirm the full rated output under the ambient outside temperature at 55 Deg C.
5.27	Manufacturer / Type / country of origin - Exciter	VTA	LEROY SOMER/PMG/CHINA	-	
5.28	Manual control setting	Required	Confirm	-	
5.29	Automatic control setting	Required	Confirm	-	
5.30	Steady state voltage adjuster	Required	Confirm	-	
5.31	CT Protection Class	5P20	Confirm	-	
5.32	Differential CT [Current Transformer]	As per Vendor Standard Protection Scheme	Confirm	-	
5.33	CT Measuring Class	Class 1	Class 1	-	
5.34	Voltage regulator	VTA	Confirm	-	
5.35	Cooling method	Fan Cooling	Confirm	-	
5.36	Installation method	IM B3	Confirm	-	
5.37	Bearing structure	VTA	Duplex bearing	-	
5.38	Overall weight	VTA	3630	-	
5.39	Moment of inertia	VTA	32.1kg.m2	-	
5.40	Parallel operation	Required	Confirm	-	

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5.41	Transient voltage adjustment rate	VTA	-35.00%	-	
5.42	Overload Capability	VTA	150% 30S	-	
5.43	Over current Capability	VTA	150% 30S	-	
5.44	Short-circuit current capability	VTA	300% 10S	-	
5.45	Voltage Variation	±10%	±10%	-	
5.46	Frequency Variation	±5%	±5%	-	
5.47	Terminal Boxes Separate terminal boxes shall be provided for: 1. Stator connections and Stator star point connections 2. Heater connections 3. Exciter connections 4. RTD connections 5. CT connection 6. Other connections	VTA	Confirm	-	
5.48	Anti-condensation Heaters	Required	Confirm	-	
6.0	GENERATOR CONTROL PANEL				
6.1	Dimension	VTA	W210*H246*D182	-	
6.2	Generator Breaker Make	VTA	schneider	-	
6.3	Generator Breaker Type	ACB [Motorized]	ACB	ACB shall be motorized	Confirmed
6.4	Generator Breaker Rating	Min 1250Amp	1250Amp	-	
6.5	Breaking Capacity	VTA	50KA	-	
6.6	Breaker Closing Time	VTA	50ms	-	
6.7	Breaker Tripping Time	VTA	50ms	-	
6.8	Remote Circuit Breaker On/Off Command and Interface	Required	Confirm	-	
6.9	Automatic Starting and Synchronization	Required	Confirm	-	
6.10	Raise/Lower Voltage Regulator	Required	Confirm	-	
6.11	Voltage Input Interface from Switchgear for Synchronization	Required	Confirm	-	
7.0	RESISTANCE & REACTANCE				
7.1	Synchronous (Xd) - Sat & Unsat	VTA	138&167	-	
7.2	Transient (X'd)	VTA	15.1&17.8	-	
7.3	Sub transient (X''d)	VTA	8.5&10	-	
7.4	Time constant of transient component (Td')	VTA	0.124	-	
7.5	Subtransient reactance time constant (Td'')	VTA	0.006	-	

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7.6	Quadrature axis synchronous reactance (Xq)	VTA	71&85	-	
7.7	Quadrature axis super transient reactance (Xq")	VTA	9.7&11.4	-	
7.8	Stator winding DC resistance (r1)	VTA	1.4	-	
7.9	Time constant direct current (Ta)	VTA	0.025	-	
7.10	Negative Sequence Reactance (X2)	VTA	9.1&10.7	-	
7.11	Parallel Operation	Required	Confirm	-	
8.0	PHYSICAL CHARACTERISTICS				
8.1	GEG Set Container Dimensions (L x W x H) in mm	VTA	12000×3000×3000	-	
8.2	Total Weight of the GEG Set Container (kg)	VTA	8500	-	
8.3	Control Panel Dimensions (L x W x H) in mm	VTA	800×800×2200	-	
8.4	Control Panel IP Rating	VTA	IP 23	-	
8.5	Dimensions (L x W x H) in mm	VTA	5125×1683×2483	-	
8.6	Weight of Alternator (kg)	VTA	3630	-	
8.7	Weight of Engine (kg)	VTA	7257	-	
8.8	Overall Weight (kg)	VTA	12500	-	
8.9	Language on Labels, Manuals and Booklets	VTA	English	-	
9.0	DEGREE OF PROTECTION				
9.1	Generator / Alternator	IP 23	Confirm	-	
9.2	Main Terminal Box	IP 23	Confirm	-	
9.3	Control Panel	IP 23	Confirm	-	
9.4	Generator Housing	IP 55 with TEFC enclosure (containerized)	Confirm	-	
10.0	CABLE DETAILS				
10.1	Power Connection cables Internal Cables: Cu/XLPE/PVC/PVC	XLPE Insulated, Copper Conductor	Confirm	-	
10.2	Cable entry	Bottom	Confirm	-	
11.0	PROTECTION				
11.1	Over current - voltage restraint (51/51V)	Required	Confirm	-	
11.2	Instantaneous Phase Over current (50)	Required	Confirm	-	
11.3	Stator Earth fault (With Neutral CT) 51N	Required	Confirm	-	

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11.4	Under voltage (27)	Required	Confirm	-	
11.5	Over voltage (59)	Required	Confirm	-	
11.6	Under frequency/Over frequency (81)	Required	Confirm	-	
11.7	Negative phase sequence (46)	Required	Confirm	-	
11.8	Reverse Power (32)	Required	Confirm	-	
11.9	Stator winding temperature (26)	Required	Confirm	-	
11.10	Bearing temperature (26)	Required	Confirm	-	
11.11	Phase sequence voltage relay (47)	Required	Confirm	-	
11.12	Local Emergency Stop (In Control Panel)	Required	Confirm	-	
11.13	Trip Supervision Relay	Required	Confirm	-	
11.14	Lockout Relay	Required	Confirm	-	
11.15	Control Circuit Supervision	Required	Confirm	-	
11.16	Other Protections	As per Vendor Standard Protection Scheme	Confirm	-	
11.17	The generator protection system and trip logic shall be in line with IEEE guidelines IEEE C37.101 and IEEE C37.102.	Compliance Required	Confirm	-	
12.0	ALARMS				
12.1	Stator winding temperature - high	Required	Confirm	-	
12.2	AVR [Auto Voltage Regulator] failure	Required	Confirm	-	
12.3	Control panel general	Required	Confirm	-	
12.4	Hardwired alarms - DG set minor trouble, majortrouble, battery charger trouble	Required	Confirm	-	
12.5	Bearing temperature - high	Required	Confirm	-	
12.6	Lock-out general (86)	Required	Confirm	-	
12.7	Under voltage relay (27)	Required	Confirm	-	
12.8	Under frequency relay (81)	Required	Confirm	-	
12.9	Over current Relay (51V)	Required	Confirm	-	
12.10	Starting battery voltage low	Required	Confirm	-	
12.11	DC power failure	Required	Confirm	-	
12.12	Incomer CB fault trip relay (86-1)	Required	Confirm	-	
12.13	Fuel low alarm level	Required	Generally, GEG is not configured	-	
13.0	METERING				

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13.1	Voltmeter	Required	Confirm	-	
13.2	Frequency	Required	Confirm	-	
13.3	kWH meter	Required	Confirm	-	
13.4	pf Meter	Required	Confirm	-	
13.5	Ammeter	Required	Confirm	-	
13.6	Running hour meter	Required	Confirm	-	
13.7	kW Meter	Required	Confirm	-	
13.8	kVAr Meter	Required	Confirm	-	
13.9	kVAH Meter	Required	Confirm	-	
13.10	kVA Meter	Required	Confirm	-	
14.0	AC POWER DETAILS				
14.1	AC System Voltage	Required	415V	-	
14.2	Variation in Supply	Required	VTA	-	
14.3	Short Circuit Level	Required	VTA	-	
15.0	BATTERY CHARGER				
15.1	Battery charger	Required	Confirm	-	
15.2	Make	VTA	Newsmy	-	
15.3	Model	VTA	C90pro	-	
15.4	Input Voltage	230VAC	AC110V-250v	-	
15.5	Float Voltage Range	VTA	12V/24V	-	
15.6	Boost Voltage Range	VTA	12V : 2V-14.5V ; 24V : 18V-29V	-	
15.7	Float charge current	VTA	0-20A	-	
15.8	Boost charge current	VTA	0-20A	-	
16.0	BATTERY				
16.1	Make	VTA	CAMEL	-	
16.2	Model	VTA	6-QW-200	-	
16.3	Nominal voltage	12V	12V	-	
16.4	Nominal capacity	Vendor shall size	200Ah	-	
16.5	Dimension of Rack	VTA	509*267*215*242mm	-	
17.0	F&G SYSTEM			Quantities shall be included in the offer	

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17.1	Flammable Gas Detector	Required	Confirm	-	
17.2	Flame Detector	Required	Confirm	-	
17.3	Temperature Detector	Required	Confirm	-	
17.4	F&G Control Panel alongwith Interface with Plant Control System & Gas Generator Control Panel	Required	Confirm	-	
18.0	Local Control Panel [Located inside the Generator Container]	Required			
18.1	Control, Protection, Metering, Interface: Electrical and Control System [Hardwired, RS485 & CAN bus]	Required	Confirm	-	
18.2	Gas Engine Driven Generators shall be interfaced to DCS and ESD system for monitoring, control and shutdown functions. Vendor shall provide hardwired and serial link interface signals for monitoring, control and shutdown functions. Interface with ESD shall be hardwired via Interposing Relay Panel. Redundant MODBUS interfacing shall be utilized for the serial communication signals.	Required	Confirm	Read/Write modbus communication to provide	
19.0	Remote Control Panel [Located inside the Package Container]	Required			
19.1	Control, Interface: Control System Required (including control and monitoring equipment and engine measurements/controls) for all the components inside the Gas Generator Enclosure alongwith external fans through the HMI installed on the panel. Vendor shall furnish details, the panel shall be installed inside the substation. Vendor shall provide Redundant serial communication (RS485) for interface from Gas Generator Vendor provided LCP to Control system (UCP).	Required	Confirm	-	
20.0	Pressure Regulating Skid	Required			
20.1	Vendor shall design as per the Fuel Gas Inlet Pressure and Requirement of the Offered Unit	VTA	Confirm	-	
21.0	Accoustic Enclsoure	Required	Confirm	-	
22.0	AC Distribution Board [For Generator Package Internal Power Distribution]	Required	Confirm	-	
22.1	Form 2 IP42 415VAC 50Hz 10kAmp 1sec 3P+4W IEC 61439 RAL 7035	VTA	Confirm	-	
22.2	Panel Rated Current shall be as per the load requirement of the Generator Auxiliaries	VTA	Confirm	-	
23.0	Fire Protection System	Required			
23.1	Required (automatic fire protection system using CO2 as the extinguishing medium)	Required	Confirm	-	
24.0	Fuel Gas Composition	Compliance Required	Confirm	-	
24.1	Component	Fuel Gas (mole fraction)		Why compliance to the composition has not been confirmed.	Compliance Confirmed
24.2	H2O	0.0000			Compliance Confirmed
24.3	Nitrogen	0.0024			Compliance Confirmed

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24.4	CO2	0.0575			Compliance Confirmed
24.5	H2S	0.00005			Compliance Confirmed
24.6	Methane	0.8290			Compliance Confirmed
24.7	Ethane	0.0942			Compliance Confirmed
24.8	Propane	0.0143			Compliance Confirmed
24.9	n-Butane	0.0011			Compliance Confirmed
24.10	i-Butane	0.0008			Compliance Confirmed
24.11	n-Pentane	0.0001			Compliance Confirmed
24.12	i-Pentane	0.0001			Compliance Confirmed
24.13	n-Hexane	0.0000			Compliance Confirmed
24.14	Total	1			Compliance Confirmed
25.0	Emission Monitoring System				
25.1	PEMS for Exhaust	Optional Price shall be provided	Optional Price shall be provided	-	
26.0	Inspection And Testing				
26.1	Insulation Resistance Tests	Required	Confirm	-	
26.2	Hydrostatic Testing	Required	Confirm	-	
26.3	Load Test	Required	Confirm	-	
26.4	Dynamic Load Test	Required	Confirm	-	
26.5	FAT: Factory Acceptance Test	Required	Confirm	-	
26.6	SAT: Site Acceptance Test	Required	Confirm	-	
27.0	SPARE PARTS & SPECIAL TOOLS				
27.1	Pre-commissioning/Commissioning/Start up spares Vendor to provide, Start-up & Commissioning Spares: As recommended by Vendor with item wise quantity and price break up (To be ordered with the equipment). As a minimum shall include 200% gaskets & 10% bolting (minimum 2 nos. Bolts/4 nuts per size) shall be provided per gasketed internal & external joints (for nozzles with B/F).	Required	Confirm	-	
27.2	Two Years Operation Spares	Optional Price shall be provided	Confirm	-	
27.3	Special Tools and Tackles for Erection, Pre-commissioning, Commissioning & Operation (if applicable)	Vendor to Furnish	No Special Tools and Tackles for Erection, Pre-commissioning, Commissioning & Operation (if	-	

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	ADNOC DOCUMENT NO: DOD-30201-50150-BH-000-16-00-004 CONTRACTOR DOCUMENT NO: ADN-AEC-EL-DAT-009	Rev.: A			
DATASHEET - GAS POWER GENERATOR					
Sr. No.	DESCRIPTION	Project Requirements	Data Filled By Vendor	ASTRA Response 20241107	MOM 20241107
27.4	A list of special tools necessary to dismantle the GEG	Vendor to Furnish	Confirm	-	
28.0	OTHER REQUIREMENTS				
28.1	Guarantee / Warranty (Months)	12 Months after commissioning	Confirm	-	
28.2	Training Manual & Installation Operation, Maintenance Manuals	Required	Confirm	-	
28.3	Supervision for Commissioning at site	Per diem rates	Confirm	-	
28.4	Delivery location	Construction Site, UAE	FCA Dongying China	Procurement action	
28.5	Delivery Duration from Purchase Order placed date	VTA	32 weeks	Agreed delivery is 27weeks as per LOA.	
28.6	Fabrication / Inspection & Test Location	Vendor to Furnish	No. 267, Nan'er Road, Dongying City, Shandong ,China.	-	
28.7	Local Agent Details	Vendor to Furnish		Local services availability shall be confirmed	KEURI shall manage by themselves
28.8	Vendor's Engineering & Procurement Location	Vendor to Furnish	No. 267, Nan'er Road, Dongying City, Shandong ,China.	-	
28.9	Preservation	Preservation suitable for 6 months storage in marine, hot and humid saline atmosphere	Confirm	-	
28.10	Vendor shall supply Gas Engine Generator (including accessories) in assembled condition. Bidder to confirm.	Compliance Required	Confirm	-	
28.11	Painting As stated above, Before shipment, all equipment's and components of the unit shall be Facility Painted to international standards, marine grade suitable for humid, dusty and saline site conditions	Required	Confirm	-	
29.0	MISCELLANEOUS				
29.1	Foundation Load Data	VTA	Confirm	-	
29.2	Installation/fixing detail with Foundation	VTA	Confirm	-	

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Sr. No.	DESCRIPTION	Project Requirements	Data Filled By Vendor	ASTRA Response 20241107	MOM 20241107
29.3	Documents Required with Offer Filled and Complied this Data Sheet Detailed Vendor Standard Datasheet of all components Vendor & Sub-Vendor List with port of origin Heat & Material Balance Diagram PID Equipment List Power Auxiliary List Consumption List Equipment Vendor Standard Catalogs MTC and Certificates Once technical closure is recieved signed stamped this datasheet	Vendor shall submit with Offer	Heat & Material Balance Diagram Equipment Vendor Standard Catalogs MTC and Certificates will be provided later. What is Vendor Standard Catalogs MTC and Certificates?	Vendor standard catalogs shall be provided for reference of replacement parts in future and also for operation & maintenance. MTC shall be part of MRB and shall be required for QAQC purpose.	Available Vendor catalogs shall be submitted by tomorrow 20241108.
29.4	Anchor Bolt	Required	The container does not need to be secured.	Noted. Further shall be discussed during detailed design	-
29.5	Painting: As per Vendor Standard	VTA	Confirm	-	
29.6	Internal and External Earthing Container shall be provided with internal earthing with suitably designed earth bars alongwith earthing cables, point for generator star earthing, external enclosure earthing ensuring safe and reliable operation	Required	Confirm	-	
29.7	Auxiliary Load, kW	Required	Confirm	-	
	VTA: Vendor to Advise [With Offer]				

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DATASHEET - GAS POWER GENERATOR					
Emissions Data For Compliance					
Emission Limits for Combustion Unit below 10MW Total thermal input					
Sr. No.	Common Name	Chemical Symbol	Liquid Fuel	Fuel Gas or Natural Gas	MOM 20241107
			Limit Value (mg/Nm3)	Limit Value (mg/Nm3)	
1	Oxides of Nitrogen	Nox	200	100 (Natural Gas) 200 (Fuel Gas)	Compliance Confirmed
2	Particulate Matter	PM	30	10	Compliance Confirmed
3	Sulphur Dioxide	SO2	350	35	Compliance Confirmed
4	Cabon Monoxide	CO	50	50	Compliance Confirmed with "Additional Catalytic convertor"

Note: Reference Conditions 273K, dry, 101.3kPa and 3% O2

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